

Literature-implied answer for arXiv:2412.20282

Defective logarithmic Sobolev inequalities survive bounded ground-state perturbations

Status. `literature_implied_answer (full bounded-perturbation case)`. This is an agent-identified composition of published theorems, not a new proof produced by the run.

Source question. In Remark 6.19, PDF pages 69–70, Gross asks whether, for a bounded potential V_1 , the ground-state measure of $-\Delta + V_0 + V_1$ satisfies a defective logarithmic Sobolev inequality whenever the ground-state measure of $-\Delta + V_0$ does. He concludes:

As to whether perturbation of a DLSI yields another DLSI under some conditions on the perturbing potential is still an open question.

Write the original ground-state probability measure as μ and its Dirichlet form as $\mathcal{E}$. Normalize the DLSI as

$$\text{Ent}_\mu(f^2) \leq 2C \mathcal{E}(f, f) + D \|f\|_{L^2(\mu)}^2. \quad (1)$$

Claim. *Assume that the symmetric Markov semigroup associated with $(\mathcal{E}, \mu)$ is ergodic. If (1) holds and W is a bounded real potential, then the ground-state measure obtained by perturbing with W satisfies a logarithmic Sobolev inequality with zero defect. In particular, it satisfies a DLSI. The same conclusion holds for an unbounded W whenever Gross’s exponential-integrability hypotheses hold with respect to the tightened LSI constant.*

In the Schrodinger setting of the source, uniqueness of the original ground state gives the required ergodicity after the ground-state transform.

Derivation. A DLSI is equivalent to hyperboundedness of the associated Markov semigroup. Consequently, for some $T > 0$ and $p > 2$, the self-adjoint Markov operator P_T is bounded from $L^2(\mu)$ to $L^p(\mu)$. Ergodicity of the semigroup implies ergodicity of P_T by spectral calculus. Miclo’s Theorem 1 [2] therefore gives a spectral gap. Equivalently, there is a finite constant C_P such that

$$\text{Var}_\mu(f) \leq C_P \mathcal{E}(f, f).$$

Gross quotes the Rothaus tightening statement as Proposition 7.16, PDF pages 85–86 [1]: combining this Poincare inequality with (1) gives

$$\text{Ent}_\mu(f^2) \leq 2c_0 \mathcal{E}(f, f), \quad c_0 = C + C_P \left(\frac{D}{2} + 1 \right). \quad (2)$$

Now apply Gross’s main perturbation theorem, Theorem 2.2, PDF page 10, to the zero-defect LSI (2) and the potential W . If W is bounded, then for every $\kappa > 0$ and every $\nu > 2c_0$,

$$\|e^W\|_{L^\kappa(\mu)} < \infty, \quad \|e^{-W}\|_{L^\nu(\mu)} < \infty.$$

The theorem produces a unique positive ground state for the perturbed operator and a zero-defect LSI for its ground-state measure. By the consecutive ground-state transformation described in Gross’s Section 8.1, this is exactly the ground-state measure of $-\Delta + V_0 + W$. This proves the claim. The last step is unchanged for unbounded W if the two displayed exponential moments are finite for some $\kappa > 0$ and $\nu > 2c_0$.

Why this is a literature implication. Miclo proves the decisive hyperboundedness-to-gap theorem but predates Gross’s question and does not claim to answer it. Gross himself cites Miclo immediately after Remark 6.19 and separately records both Rothaus tightening and the required perturbation theorem, but does not compose the three facts. The relation to the stated question is therefore an identification, not an independent new mathematical ingredient.

Limitations. Ergodicity is essential to Miclo’s theorem, so the argument does not cover arbitrary reducible forms. The conversion from (1) to a gap is qualitative: Miclo’s Proposition 11 shows that no lower gap bound depending only on a general hyperboundedness norm can exist. Thus c_0 is explicit only when a Poincare constant C_P is independently available.

Search evidence. Cheap run indexes and bounded web/arXiv searches through 2026-08-09 used arXiv:2412.20282, the exact question sentence, and combinations of “defective logarithmic Sobolev”, “bounded perturbation”, “ground state”, and “Miclo”. No later paper explicitly making this identification was found.

References

- [1] L. Gross, *Invariance of intrinsic hypercontractivity under perturbation of Schrodinger operators*, Advanced Nonlinear Studies 25 (2025), no. 4, 951–1024; arXiv:2412.20282v2.
- [2] L. Miclo, *On hyperboundedness and spectrum of Markov operators*, Inventiones Mathematicae 200 (2015), no. 1, 311–343, doi:10.1007/s00222-014-0538-8.